

ENGR 1451/2451: Exploratory Data Science

Measurement Error

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Today's Goals

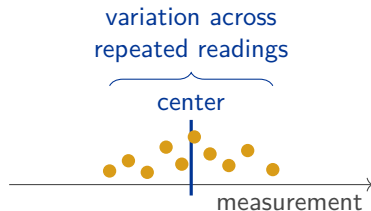
Understand what repeated measurements reveal—and what they cannot reveal.

- Explain chance error and why repeated readings vary.
- Summarize readings with the sample mean $\bar{x}$ and sample standard deviation s .
- Distinguish chance error, bias, precision, and accuracy.
- Evaluate outliers and the limits of a normal approximation.

Python companion: `w04a_measurement_error.ipynb`

Why Repeat a Measurement?

- Even under controlled conditions, repeated readings vary.
- This uncontrolled variation is called **chance error**.
- One reading cannot reveal the size of that variation.
- Repetition lets us estimate the **center** and **spread** of the measurement process.



Is One Measurement Enough?

True or false? Explain.

An experienced scientist using the best equipment needs to measure an object only once. Without a mistake, a second reading will be identical.

Discuss: Is chance error the same as a mistake?

Chance Error: A Model for One Reading

$$\text{measurement} = \text{exact value} + \text{chance error}.$$

- The **exact value** is fixed but usually unknown.
- **Chance error** changes from reading to reading.
- When chance errors balance around zero, the average of many readings estimates the exact value.

An important assumption

This model has no systematic bias. Repeating a flawed procedure does not remove bias.

Repeated Weighings: The NB 10 Example

The Bureau weighed the same nominal 10-gram standard 100 times. Each reading is the **shortfall below 10 grams**, in micrograms:

409, 400, 406, 399, 402, ...

Sample mean: center

$$\bar{x} \approx 405 \mu\text{g}$$

Estimates the exact shortfall **if the procedure is unbiased**.

Sample standard deviation: spread

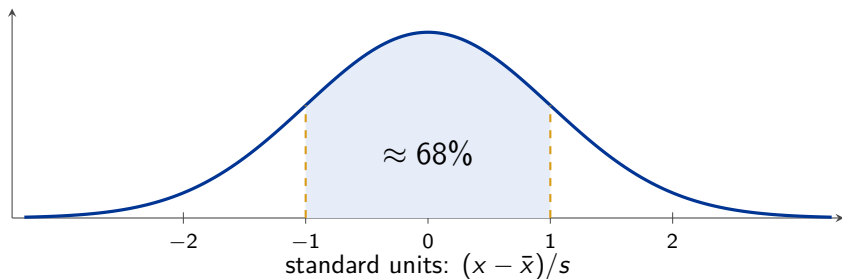
$$s \approx 6 \mu\text{g}$$

Estimates the population standard deviation σ of chance error in a **single reading**.

These are rounded summaries. The spread of individual readings is not the uncertainty of their average.

One Standard Deviation and the Normal Curve

For an approximately normal distribution of readings:



About 68% fall between $\bar{x} - s$ and $\bar{x} + s$.

This percentage is not a rule for every dataset.

Example: A Fourth Calibration

A laboratory calibrates the same yardstick three times:

35.96 in. 36.01 in. 36.03 in.

What would you expect next?

A fourth reading should be about 36 inches, give or take:

A. 0.01 in. **B.** 0.03 in. **C.** 0.06 in.

Discuss: Which feature of the readings determines your choice?

Chance Error and Bias Are Different

Chance error

- Varies from reading to reading.
- Creates spread, estimated by s .
- Can partly balance out when readings are averaged.

Bias

- Pushes readings systematically high or low.
- Does not disappear with repetition.
- Requires calibration or an independent reference to assess.

measurement = exact value + bias + chance error.

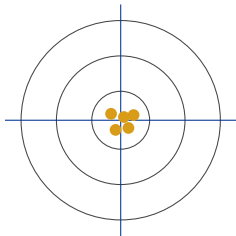
A Tape Measure: Random Error vs. Bias

A carpenter repeatedly measures the length of a board.

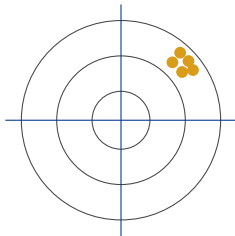
1. What could cause **bias**?
2. What could cause **random error**?
3. Which is more susceptible: a **steel tape** or a **cloth tape**?
4. Could the cloth tape's bias change over time?

Precision Is Not the Same as Accuracy

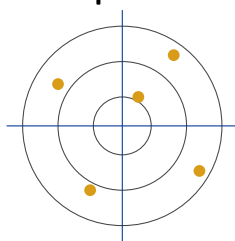
Precise and accurate



Precise but biased



Imprecise



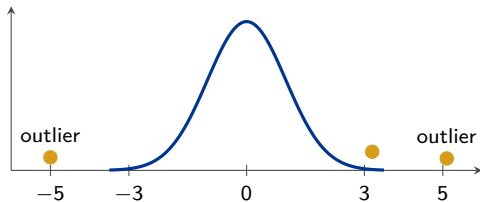
Target center = exact value.

Course terminology

Precision: small chance-error spread (small s).

Accuracy requires small bias relative to the correct value. A tight cluster can still be systematically wrong.

Outliers Can Strongly Affect s



Illustration, not the raw NB 10 data.

NB 10: rounded summaries

Readings	$\bar{x}$	s
All 100	405	6
Without 3 extremes	404	4

Units: micrograms below 10 g.

The mean changes little; s decreases by about **30%**.

Investigate; do not delete automatically

An outlier may be a blunder, a rare chance error, or evidence of an incomplete model. A better normal fit is not, by itself, a reason to discard it.

Theranos: Excluding Results to Pass a Test

Reported in trial testimony

Former lab associate Erika Cheung testified that selected **quality-control results** were removed as “outliers” so the remaining results would pass.

She described a lack of rules for choosing which points to exclude.

The statistical lesson

An exclusion needs a **scientifically justified rule** or an **identified measurement failure**—not merely a better-looking result.

Do not choose exclusions simply to make a quality check pass.

What a Normal Fit Can and Cannot Show

If the fit is adequate, it can describe

- The center of repeated readings.
- Chance-error spread.
- Approximate percentages within s , $2s$, or $3s$ of $\bar{x}$.
- Unusually distant observations.

It cannot establish

- That the instrument is unbiased.
- That the center is the exact value.
- That an outlier is a mistake.
- That future conditions will match calibration conditions.

A tight, normal-looking distribution may be **repeatable but biased**.

Example: Is the Normal Estimate Reliable?

100 readings on NB 10

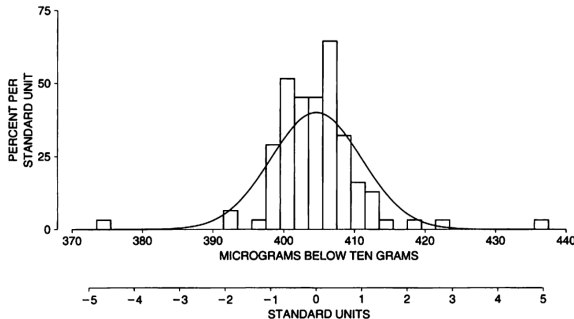
Use the chapter's rounded summaries:

$$\bar{x} \approx 405 \mu\text{g}, \quad s \approx 6 \mu\text{g}.$$

Consider readings from **400 to 406 micrograms**, including both endpoints ($400 \leq x \leq 406$).

Is the normal estimate **too low, too high, or about right?**

Explain using the histogram.



Reporting Outliers: Reject Only for Cause

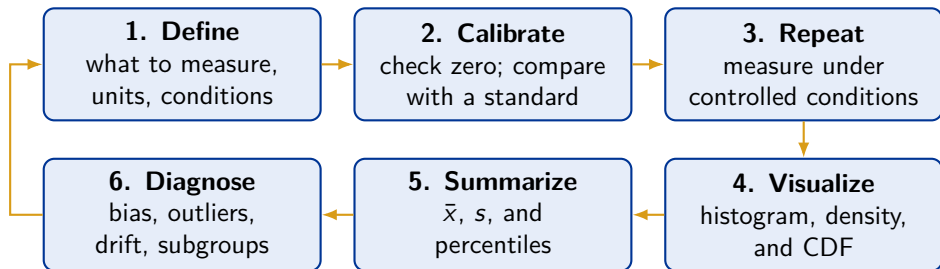
National Bureau of Standards, quoted in the chapter

“Rejection of data on the basis of arbitrary performance limits severely distorts the estimate of real process variability. . . . Realistic performance parameters require the acceptance of all data that cannot be rejected for cause.”

Course takeaway: Describe actual performance, not desired performance.

- Investigate unusual observations before excluding them.
- Explain exclusions and report how they affect the summaries.

A Practical Measurement Workflow



Report the results and the conditions

Report the estimated center and spread. State how calibration, outliers, and measurement conditions were handled.

Summary

- **Repeat** measurements to reveal chance-error variability.
- **Center:** $\bar{x}$ estimates the exact value when the procedure is unbiased.
- **Spread:** s estimates chance-error variation in a single reading.
- **Bias** does not disappear merely by repeating the procedure.
- **Outliers** require investigation, not automatic deletion.
- A **normal-looking distribution** is not proof of accuracy.

Measurement Error